

Intervals and absolute value equations

Two-vertex certification of nonsingular sign regularity

Difficulty: challenging **Importance:** interesting to specialist**Status:** Partially resolved**Area:** structured matrices and interval linear algebra**Rating rationale:** Challenging reflects extending two-vertex certification through zero minors in arbitrary dimension; specialist impact is the efficient recognition of structured interval matrix families.**Further partial resolution — 12 September 2026****Author:** Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation.

[Theorem G](#) proves the full interval conclusion when the directed graph of nonzero fixed entries is acyclic and fixed zero entries occupy at most one checker parity. Its scaling argument uses Adm–Garloff’s Theorem 3.3 with its original attribution. [Theorem V](#) proves the full conclusion in dimension five if there are no fixed zeros and every fully fixed adjacent two-by-two block is nonsingular, allowing cyclic fixed graphs. For larger dimensions, Theorem V controls order three only.

The [order-two reduction](#) and the nonsingularity/cofactor lemma control orders one, two, and the two highest orders under the original hypotheses. The unrestricted remaining middle-minor signs are not settled. Neither sufficient condition above covers every interval in the original question, which remains unchanged below.

These partial arguments passed a separate [independent Codex AI-agent review](#). [Submission record](#), [verified affiliation and exact certificates](#). AI assistance is disclosed; finite certificates support examples rather than prove the universal theorem. No historical priority, external human peer review or formal verification is claimed. No Lean verification was performed.

Problem statement

For $\epsilon \in \{-1, 1\}^n$, call a real $n \times n$ matrix M **nonsingular sign regular with signature ϵ** if $\det M \neq 0$ and every minor of order k has determinant d satisfying $\epsilon_k d \geq 0$, for every $1 \leq k \leq n$.

Let $n \geq 5$ and let $L, U \in \mathbb{R}^{n \times n}$ satisfy $L \leq U$ entrywise. Define the two checkerboard vertex matrices

$$A_{ij}^- = \begin{cases} L_{ij}, & i + j \text{ even,} \\ U_{ij}, & i + j \text{ odd,} \end{cases} \quad A_{ij}^+ = \begin{cases} U_{ij}, & i + j \text{ even,} \\ L_{ij}, & i + j \text{ odd.} \end{cases}$$

Question

If A^- and A^+ are nonsingular sign regular with the same signature ϵ , must every real matrix M with $L \leq M \leq U$ be nonsingular sign regular with signature ϵ ?

The implication would certify a property of an entire interval of matrices from only two of its vertices. Allowing zero minors is essential to the remaining problem.

References

Jürgen Garloff, Mohammad Adm, and Jihad Titi, *A Survey of Classes of Matrices Possessing the Interval Property and Related Properties*, Reliable Computing **22** (2016), 1–14, Conjecture 3.1, p. 7. Adm and Garloff, *Certification of the Sign Regularity of Matrix Intervals*, Acta Scientiarum Mathematicarum, published January 17, 2026, §3, especially the question following Theorem 3.2 and the discussion of admissible signatures.

Earlier status evidence — 2026-09-08

The 2026 survey explicitly retains this question. It records positive results for strictly sign regular matrices, totally nonnegative matrices, tridiagonal matrices, and certain signatures covering all orders at most four. Theorem 3.3 also covers intervals whose fixed entries $L_{ij} = U_{ij}$, if any, all have the same parity of $i + j$; fixed entries of both parities are allowed in the general question. It also records counterexamples when nonsingularity is omitted. Searches for sign-regular interval conjectures and subsequent 2026 results found no resolution of the displayed implication.

Audit update — 2026-09-10

Rechecked the [2026 survey](#), §3, Theorems 3.2–3.3 and the intervening open question. Tridiagonal and certain signature classes, as well as the specified restrictions on fixed-entry parity, prove substantive subclasses even for $n \geq 5$. The unrestricted nonsingular case remains explicitly open there. Searches for subsequent sign-regular interval certification results found no general resolution; the status now exposes these partial results.